ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)

ACADEX PRACTICE EXAMINATION

O-LEVEL · MATHEMATICS

MATHEMATICS 4004/1

PAPER 1 June 2023

2 hours 30 minutes

PAPER 1 — SHORT-ANSWER — CALCULATORS MUST NOT BE USED

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to June 2023 4004/1.

Additional materials: Electronic calculators must NOT be used. Geometrical instruments may be used.
INSTRUCTIONS: Answer all 30 questions. Show all working. The number of marks is given in brackets [ ].

1. Simplify 2^4 × 2^2. Leave as a power of 2. **[2]**

2. Find the HCF and LCM of 36 and 16. **[3]**

3. Write 7000000 in standard form. **[2]**

4. n(A) = 17, n(B) = 9, n(A ∩ B) = 5. Find n(A ∪ B). **[2]**

5. Work out 1 + 1/3. Give a mixed number if possible. **[4]**

6. Expand (x + 1)(x + 3). **[4]**

7. Express 36 as a product of its prime factors. **[4]**

8. A map of Chitungwiza uses scale 2 : 700000. A road is 10 cm on the map. Find the real length in km. (1 : 100 000 means 1 cm ↔ 1 km if 2:7 is first simplified — instead: scale 2 cm to 7 km.) **[4]**

9. Overnight in Chitungwiza the temperature was 5°C. By 14:00 it was 21°C. Find the rise. **[3]**

10. Find 30% of 250. **[3]**

11. Work out (1 1 ; 3 2)(1 ; 5). **[4]**

12. Factorise $x^2 + 6x + 5$. **[4]**

13. Factorise completely $10x + 10$. **[3]**

14. Find the mean of 4, 8, 11, 7, 11. **[3]**

15. AB ∥ CD. A transversal makes an acute angle of 50° with AB. Find the co-interior angle on CD. **[3]**

16. Make h the subject of $A = 2\pi r(r + h)$. **[4]**

17. A shirt in Chitungwiza is marked $80. In a sale it is 25% off. Sale price? **[4]**

18. A bag contains 5 red, 6 blue and 2 green beads. P(red)? **[3]**

19. Solve<<
>>x + y = 11<<
>>x − y = 1. **[4]**

20. In right-angled triangle PQR, $\hat{P} = 30°$ and hypotenuse PR = 16 cm. Find PQ, opposite 30°. **[4]**

21. A sequence is 9, 13, 17, 21, ... Find an expression for the nth term. **[4]**

22. A right-angled triangle has shorter sides 8 cm and 15 cm. Find the hypotenuse. **[4]**

23. Solve $4x + 5 = 29$. **[3]**

24. Solve $3x + 2 < 17$. **[3]**

25. A triangular plot has base 8 m and perpendicular height 7 m. Find its area. **[3]**

26. Find the midpoint of (4, 4) and (5, 5). **[3]**

27. A cuboid water tank measures 6 m by 5 m by 5 m. Find its volume. **[3]**

28. Find the gradient of the line joining (2, 1) and (6, 7). **[3]**

29. A regular polygon has 8 sides. Find each exterior angle. **[3]**

30. Circumference of a circle radius 7 cm. Take $\pi = 22/7$. **[4]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).